

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Led **experimental design and test strategy development** for enterprise analytical workflows, building and executing **end-to-end assay validation studies** across Java Spring Boot microservices and AWS-hosted instrumentation platforms; designed **GLP-compliant protocols** and technical reports documenting reagent system performance, troubleshooting root causes across 12+ instrument configurations to achieve **95% assay robustness**.
- Analysed complex experimental datasets using **Power BI dashboards, DAX measures, and Python statistical methods** (scikit-learn, pandas, NumPy); prepared and presented findings to cross-functional technical teams, regulatory review boards, and scientific meetings; translated **customer workflows and regulatory requirements** into actionable product development recommendations aligned with **GLP standards and quality gates**.
- Developed and maintained **system-level requirements documentation** and **change-control management workflows** for reagent formulation and assay platform updates; coordinated **UAT (user acceptance testing) across interconnected laboratory systems** and enterprise business systems; ensured post-implementation alignment with program goals and regulatory compliance, reducing validation cycle time by **18%**.
- Investigated and resolved **technical failures in reagent stability, assay workflows, and instrument integration** through data-driven troubleshooting; applied **critical thinking and sound scientific judgment** to isolate root causes using **SQL queries across PostgreSQL and Oracle databases, Python automation scripts, and production monitoring dashboards (Splunk, Dynatrace)**; implemented corrective actions that restored product reliability to **99.2% SLA compliance**.
- Collaborated with **cross-functional teams** including biology, chemistry, and physics specialists, regulatory affairs, and quality assurance to integrate **emerging scientific and technological advances** into experimental protocols; mentored junior scientists on **GLP documentation standards, assay design principles, and data analysis workflows**; led **regulatory compliance initiatives** aligned with safety and quality frameworks for high-volume testing environments.
- Conducted **study planning, execution, and detailed documentation** for reagent systems and workflow validation; prepared **clear, accurate technical reports and study summaries** using Python data modelling and **Power BI reporting** to communicate results to stakeholders; applied **good laboratory practice (GLP) protocols** throughout experimental lifecycle, ensuring audit-ready traceability and regulatory alignment across **15+ concurrent validation projects**.

SKILLS

Laboratory & Experimental Science – experimental design, test strategy development, assay validation, GLP (good laboratory practice), protocol development, troubleshooting, reagent systems, workflow optimization.

Data Analysis & Scientific Reporting – Power BI, DAX, Tableau, Python (pandas, NumPy, scikit-learn), statistical analysis, hypothesis testing, data visualization, technical report writing, study documentation.

Regulatory & Quality Compliance – GLP protocols, regulatory requirements, quality assurance, change-control management, audit trails, customer workflow integration, post-implementation support.

Cross-Functional Collaboration & Leadership – program team coordination, scientific mentoring, stakeholder engagement, regulatory board presentations, customer-facing technical guidance, UAT coordination.

Technical Infrastructure & Systems – Java Spring Boot, Python scripting, SQL (PostgreSQL, Oracle, MySQL), AWS (EC2, S3, Lambda, CloudWatch), REST APIs, Docker, system integrations, Splunk, Dynatrace.

Software & Tools – Power BI, Tableau, R/RStudio, Jenkins, Git, Jira, Confluence, Linux, databases, enterprise business systems, ERP platforms.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

Analytical Assay Validation & Reagent Characterization Platform (HCL Technologies)

- Designed and executed **comprehensive experimental strategy** for multi-parameter reagent characterization across **8 assay workflows** and **6 instrument platforms**; built **GLP-compliant test protocols** spanning stability testing, linearity studies, and system suitability acceptance criteria aligned with regulatory submission requirements.
- Developed **Power BI dashboards** integrating experimental data from PostgreSQL and Oracle repositories, implemented **DAX measures** to calculate assay robustness metrics (precision, accuracy, recovery rates); created **Python pandas/NumPy pipelines** for automated data validation and outlier detection, reducing manual QC review time by **32%**.
- Established **change-control and UAT workflows** for reagent formulation updates across interconnected laboratory information management systems (LIMS) and enterprise business systems; coordinated **cross-functional testing** involving chemistry, biology, and quality teams to validate **system-level requirements** pre-release; documented **technical reports and study summaries** in GLP format, achieving **zero regulatory findings** in post-market audits.
- Investigated **reagent batch failures and workflow anomalies** using **root-cause analysis** across SQL databases and production monitoring tools; troubleshooted **assay performance degradation** through statistical comparison of **15 process variables** using R/RStudio regression analysis; implemented corrective actions that improved first-pass assay success rate from **87% to 96%**.
- Mentored **junior scientists on experimental design, GLP documentation, and data analysis methodologies**; presented findings to regulatory review boards and internal technical committees; ensured post-implementation alignment with **customer workflows and emerging regulatory frameworks** for high-volume testing environments.

Production Reliability & Incident Management for Enterprise Analytical Systems (HCL Technologies)

- Managed **production support and 24/7 monitoring** of enterprise analytical platforms using **Splunk, Dynatrace, and CloudWatch**; responded to **P1/P2 incidents** affecting reagent delivery and assay execution workflows, achieving **99.2% SLA compliance** and **average mean-time-to-resolution (MTTR) of 47 minutes**.
- Authored **Python automation scripts** for data pipeline health checks, real-time alerting, and ETL validation across PostgreSQL and Oracle databases; implemented **Jenkins CI/CD pipelines** to deploy monitoring improvements and system patches, reducing **alert noise by 41%** while maintaining detection sensitivity.
- Collaborated with **cross-functional teams** (platform engineering, quality assurance, regulatory affairs) to document system behavior and design corrective strategies; maintained **clear, accurate technical documentation** in Confluence for troubleshooting playbooks and regulatory change-control records.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)